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# Daily Expense Tracker (Week 4: improved version after testing)
# Problem : People forget where their money goes every month.
# Solution: A small Python program that records each expense,
#           saves it in a file and shows a summary with a budget warning.
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import csv                      # lets us read and write CSV (spreadsheet-like) files
import os                       # lets us check if a file already exists
import math                     # lets us check that a number is a real, finite number
from datetime import date       # gives us today's date

FILE_NAME = "expenses.csv"      # the file where all expenses are saved
BUDGET = 5000                   # monthly budget in rupees (change as needed)


def load_expenses():
    """Step 1: Read old expenses from the file so nothing is lost."""
    expenses = []                               # an empty list to hold all expenses
    skipped = 0                                 # counts damaged lines in the file
    if os.path.exists(FILE_NAME):               # condition: only read if file exists
        with open(FILE_NAME, newline="") as f:  # open the file for reading
            for row in csv.reader(f):           # loop: go through each saved line
                try:
                    # each row = [date, category, amount]
                    expenses.append((row[0], row[1], float(row[2])))
                except (IndexError, ValueError):    # IMPROVEMENT 3: skip damaged lines
                    skipped += 1
    if skipped:
        print(f"Skipped {skipped} damaged line(s) in {FILE_NAME}")
    return expenses                             # give the list back to the caller


def save_expense(day, category, amount):
    """Step 2: Add one new expense at the end of the file."""
    with open(FILE_NAME, "a", newline="") as f:  # "a" means append (add at end)
        csv.writer(f).writerow([day, category, amount])


def show_summary(expenses):
    """Step 3: Show the total, the total per category and a budget message."""
    total = sum(item[2] for item in expenses)    # add up every amount
    print("\n--- Summary ---")
    print("Total spent:", total)

    totals = {}                                  # dictionary: category -> money spent
    for _, category, amount in expenses:         # loop through all expenses
        totals[category] = totals.get(category, 0) + amount
    for category, amount in totals.items():      # print each category total
        print(f"  {category}: {amount}")

    if total > BUDGET:                           # condition: compare with budget
        print("Warning: Budget limit crossed!")
    else:
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        print("Money left in budget:", BUDGET - total)


def main():
    """Step 4: The main program that talks to the user."""
    expenses = load_expenses()                  # start with old data
    print("Daily Expense Tracker (type q to quit)")

    while True:                                 # loop: keep asking until user quits
        category = input("\nCategory (Food/Travel/Bills/Fun/Other) or q: ").strip().title()
        if category == "Q":                     # user typed q -> stop the loop
            break

        # IMPROVEMENT 1: a category must be a name made of letters (spaces allowed)
        if not category.replace(" ", "").isalpha():
            print("Please enter a category name using letters only, for example Food")
            continue

        try:
            amount = float(input("Amount: "))    # convert the typed text to a number
        except ValueError:                       # runs if user typed letters
            print("Please enter a number, for example 120")
            continue                             # go back and ask again

        if not math.isfinite(amount):            # IMPROVEMENT 2: reject nan and inf
            print("Please enter a number, for example 120")
            continue

        if amount <= 0:                          # input check: no zero or negative
            print("Amount must be more than 0")
            continue

        today = str(date.today())                # today's date as text
        expenses.append((today, category, amount))   # add to the list
        save_expense(today, category, amount)        # and save to the file
        print("Saved!")

    show_summary(expenses)                       # show results after quitting


if __name__ == "__main__":                       # run main() only when file is run directly
    main()
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